



KUALA LUMPUR UNIVERSITY OF SCIENCE AND TECHNOLOGY

(formerly known as Infrastructure University Kuala Lumpur)

Project Human Resource Management in Software Projects:

Evaluating Challenges and Proposing Strategies for TechNova Solutions

Individual Assignment

Name:	Mohamed Mohamed Hussein
Student ID:	251925403
Course:	CSC 2332-SOFTWARE PROJECT MANAGEMENT
University:	Kuala Lumpur University of Science and Technology
Lecturer:	SUHAILA BINTI MOHD NORDIN
Date:	May 31, 2026

Abstract

In this paper we identify four human resources issues at TechNova Solutions in developing an LMS, specifically: communication, missed deadlines, hoarding knowledge, and conflict among team members. In addition to identifying five different management strategies for addressing these issues as a whole namely: leadership, conflict resolution, motivation, ethics, and communication we present an HR support strategy for TechNova's project team using Jira. My results show that structured HR strategies combined with digital technology can be beneficial in improving both the performance of teams and their project related outcomes.

1. Introduction

Software projects rely heavily on collaborative work between individuals. Unlike physical projects Software development demands continuous coordination of knowledge sharing and collaboration to problem solve. Therefore people management is just as important as technical task management in Software development (Schwalbe, 2019).

According to the Project Management Body of Knowledge (PMBOK), Project HRM are those processes that are required to Plan, Acquire, Develop and Manage the project team (Project Management Institute [PMI] 2021). Failure in these process can cause delays, poor quality, and conflicts among team members. This article will evaluate four HRM issues at TechNova Solutions. It will propose five potential management solutions to address each of the identified issues. In addition, it will create a software supported HRM plan for an eight month long LMS project.

2. Evaluation of Human Resource Problems

2.1 Poor Communication

Poor communication is one of the most damaging problems that a software team can experience. PMI (2021) states over half of project failures stem from poor team communication. When developers do not tell testers about changes made to modules, testers create test cases based on outdated information. As a result, developers are required to rework items that were previously tested correctly and inconsistent behavior is exhibited by the software. Additionally, poor communication among team members will also lead to decreased trust with each other which will make future collaboration more difficult

2.2 Chronic Deadline Misses by Senior Developers

Two senior developers for TechNova have difficulty meeting deadlines. The impact is significant since senior developers are responsible for the most difficult projects. When they fall behind, it blocks other people who rely on their output. According to Schwalbe (2019), when major players in a project fail to meet their deadlines, there is a ripple effect throughout the entire project. As such, when one developer misses his or her deadline during sprints, the velocity of the team decreases as well as causes additional workload for other team members. This can lead to both frustration and an unbalanced distribution of work.

2.3 Knowledge Hoarding

When some team members withhold their technical knowledge, a small group of people develop a dependence upon each other. The importance of sharing knowledge was identified by Davenport & Prusak (1998) for both teams and organizational learning. At TechNova when an employee that retains knowledge leaves or is unavailable there is a loss of key skills and no ability to quickly recover. Further, this situation prevents new employees from being able to learn best

practices and therefore creates inconsistencies in the quality of code developed throughout the project.

2.4 Conflict Between Testing and Development Teams

Developers and testers at many organizations have differing objectives; developers want to show how much has been accomplished, while testers want to identify all issues. If these competing interests are not managed effectively, they will become personal conflicts. Thomas and Kilmann (1974) reported that unmanaged conflict can lead to a loss in both team cohesion and production. When the developer and tester teams fail to work together at TechNova, some bugs go unfixed or undiagnosed for an extended period, resulting in lower final quality.

3. Strategies for Managing People Effectively

3.1 Transformational Leadership

The Project Manager needs to use a Transformational Leadership Style. Transformational Leaders are described by Bass and Riggio (2006) as leaders that have the ability to motivate Teams through Vision, Motivation and Role Modeling. For the Project Team at TechNova this will include Regular Alignment Meetings where Contributions are recognized in Public forums with connection of Daily Tasks to the overall Goal of the Project.

3.2 Structured Conflict Resolution

Establishing a formal mediation process for conflict resolution would allow TechNova's Testing Team and Development Teams to formally resolve potential disputes on issues like priority of tasks. Thomas and Kilmann (1974) recommended using a "collaborating" approach when there is an apparent conflict in priorities based on two or more legitimate competing interest areas. Scheduled cross-teams review meetings will help to eliminate the opportunity for unresolved conflicts to build by providing a regularly scheduled time for open discussion.

3.3 Fair Motivation and Workload Management

Herzberg (1959), and Maslow (1943) both concluded that employees' needs for recognition and purpose in their job. Motivation at TechNova has decreased mostly because of unfair workloads. To increase fairness and recognize employees who consistently demonstrate effort, you can implement a workload tracking system to assign tasks more evenly and develop a very basic recognition program to motivate staff.

3.4 Enforcing Ethical Standards

The leakage of confidential company project data is a severe offense against professional ethics. The Project Management Institute (PMI) Code of Ethics (2006) states that project managers have an obligation to exhibit integrity and take responsibility for their actions. All staff at TechNova should be required to sign a Non-Disclosure Agreement (NDA) as well as participate in on going ethics training. Any violations should be processed using a documented disciplinary action plan which will send a strong message to all employees of TechNova that unethical behavior will not be tolerated.

3.5 Agile Communication Practices

By adopting agile methodologies like a daily meeting called "standup", at end of each sprint review and after each sprint retrospective, teams have improved their communication (Schwaber & Sutherland, 2020). The daily meetings and other recurring communications will help to surface all issues/roadblocks quicker so the team remains in sync. To combat knowledge hoarding pair development and set times for sharing knowledge across the group will enable a stronger sense of teamwork and reduce the reliance on one or two individuals.

4. Software-Supported HRM Plan

TechNova should use Jira as its central project management platform. Table 1 outlines how Jira supports each HRM area.

The implementation timeline for this eight month LMS software development project, mapped out across its consecutive analysis, development, and testing iterations, is illustrated in Gantt chart below.

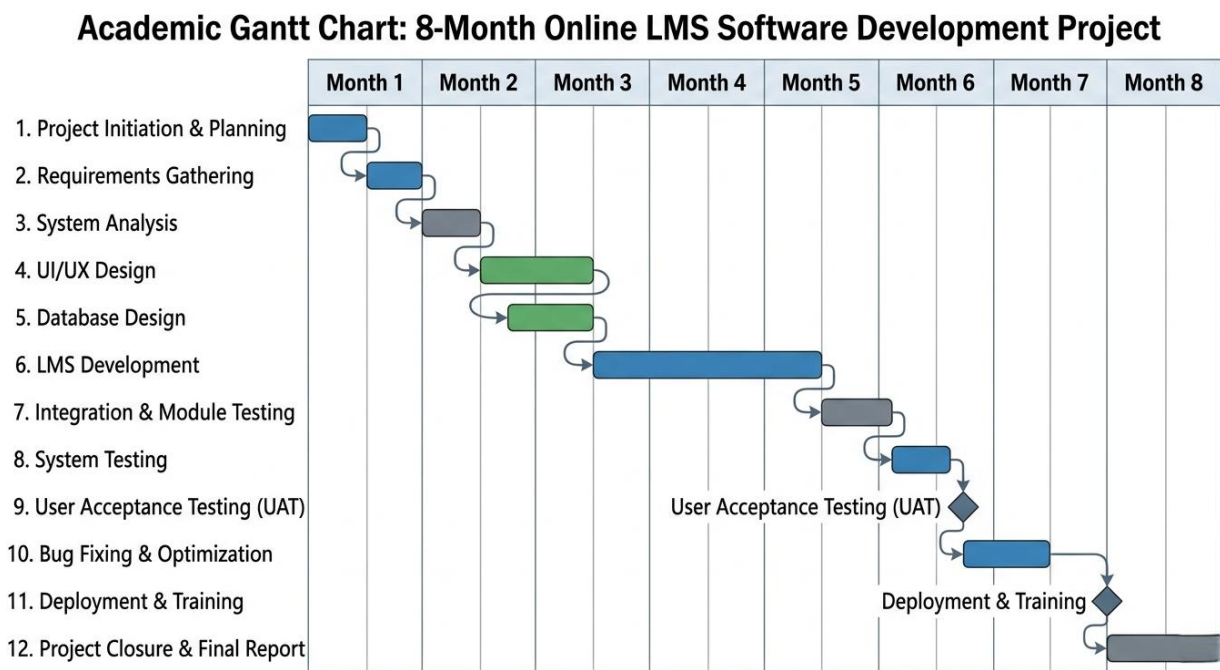


Table 1

Jira HRM Support Plan for TechNova LMS Project

HRM Area	Jira Feature	Purpose
Task Allocation	Backlog & Sprint Planning	Assign tasks with clear owners, deadlines, and priority levels
Progress Monitoring	Burndown Chart & Board	Track sprint progress and flag members falling behind
Communication	Comments & Confluence	Enable in-task discussions and shared documentation

HRM Area	Jira Feature	Purpose
Conflict & Issues	Issue Tracker	Log blockers and inter-team dependencies to flag conflicts early
Reporting	Velocity Reports & Dashboards	Provide weekly workload and output summaries to the project manager

Note: Adapted from Atlassian Jira documentation (Atlassian, 2023).

Jira's sprint planning board allows the project manager to assign user stories with clear descriptions, acceptance criteria, and due dates, removing ambiguity and reducing communication gaps. The burndown chart makes it easy to spot if a developer is consistently missing task completions, allowing early intervention before delays become critical. The workload dashboard supports fair task distribution, and Confluence enables shared documentation and knowledge-sharing, directly addressing knowledge hoarding.

5. Conclusion

The HR problems at TechNova poor communication, deadline misses, knowledge hoarding, and team conflict reflect a team lacking structured management processes and a shared professional culture. The five strategies proposed provide a practical framework for improvement. When combined with a well-implemented Jira plan, TechNova can better manage its people and improve the LMS project's chances of success. Future projects should embed these HRM practices from the start rather than applying them reactively.

References

- Atlassian. (2023). Jira software documentation. Atlassian. <https://support.atlassian.com/jira-software-cloud/>
- Bass, B. M., & Riggio, R. E. (2006). *Transformational leadership* (2nd ed.). Lawrence Erlbaum Associates.
- Davenport, T. H., & Prusak, L. (1998). *Working knowledge: How organizations manage what they know*. Harvard Business School Press.
- Hackman, J. R., & Oldham, G. R. (1976). Motivation through the design of work: Test of a theory. *Organizational Behavior and Human Performance*, 16(2), 250–279. [https://doi.org/10.1016/0030-5073\(76\)90016-7](https://doi.org/10.1016/0030-5073(76)90016-7)
- Herzberg, F. (1959). *The motivation to work* (2nd ed.). John Wiley & Sons.
- Maslow, A. H. (1943). A theory of human motivation. *Psychological Review*, 50(4), 370–396. <https://doi.org/10.1037/h0054346>
- Project Management Institute. (2006). *PMI code of ethics and professional conduct*. PMI. <https://www.pmi.org/about/ethics/code>
- Project Management Institute. (2021). *A guide to the project management body of knowledge (PMBOK guide)* (7th ed.). PMI.
- Schwalbe, K. (2019). *Information technology project management* (9th ed.). Cengage Learning.
- Schwaber, K., & Sutherland, J. (2020). *The Scrum guide*. Scrum.org. <https://scrumguides.org/>
- Thomas, K. W., & Kilmann, R. H. (1974). *Thomas-Kilmann conflict mode instrument*. Xicom.